Chapter Review

Key Terms

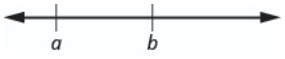
coefficient The constant that multiplies the variable(s) in a term is called the coefficient.
composite number A composite number is a counting number that is not prime.
divisibility If a number m is a multiple of n, then we say that m is divisible by n.
equation An equation is made up of two expressions connected by an equal sign.
evaluate To evaluate an algebraic expression means to find the value of the expression when the variable is replaced by a given number.
expressions An expression is a number, a variable, or a combination of numbers and variables and operation symbols.
least common multiple The smallest number that is a multiple of two numbers is called the least common multiple (LCM).
like terms Terms that are either constants or have the same variables with the same exponents are like terms.
multiple of a number A number is a multiple of n if it is the product of a counting number and n.
prime factorization The prime factorization of a number is the product of prime numbers that equals the number.
prime number A prime number is a counting number greater than 1 whose only factors are 1 and itself.
solution of an equation A solution to an equation is a value of a variable that makes a true statement when substituted into the equation. The process of finding the solution to an equation is called solving the equation.
term A term is a constant or the product of a constant and one or more variables.

Key Concepts

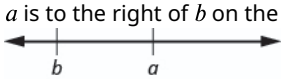
2.1 Use the Language of Algebra

Operation	Notation	Say:	The result is...
Addition	$a + b$	a plus b	the sum of a and b
Multiplication	$a \cdot b, (a)(b), (a)b, a(b)$	a times b	The product of a and b
Subtraction	$a - b$	a minus b	the difference of a and b
Division	$a \div b, a/b, \frac{a}{b}, b\overline{)a}$	a divided by b	The quotient of a and b

- **Equality Symbol**
  - $a = b$ is read as a is equal to b
  - The symbol = is called the equal sign.
- **Inequality**
  - $a < b$ is read a is less than b
  - a is to the left of b on the number line



  - $a > b$ is read a is greater than b
  - a is to the right of b on the number line



Algebraic Notation	Say
$a = b$	a is equal to b
$a \neq b$	a is not equal to b
$a < b$	a is less than b

Table 2.14

Algebraic Notation	Say
$a > b$	a is greater than b
$a \leq b$	a is less than or equal to b
$a \geq b$	a is greater than or equal to b

Table 2.14

- **Exponential Notation**
  - For any expression a^n is a factor multiplied by itself n times, if n is a positive integer.
  - a^n means multiply n factors of a

    base → a^n ← exponent

    $a^n = \underbrace{a \cdot a \cdot a \cdot \ldots \cdot a}_{n \text{ factors}}$
  - The expression of a^n is read a to the nth power.

Order of Operations When simplifying mathematical expressions perform the operations in the following order:

- Parentheses and other Grouping Symbols: Simplify all expressions inside the parentheses or other grouping symbols, working on the innermost parentheses first.
- Exponents: Simplify all expressions with exponents.
- Multiplication and Division: Perform all multiplication and division in order from left to right. These operations have equal priority.
- Addition and Subtraction: Perform all addition and subtraction in order from left to right. These operations have equal priority.

2.2 Evaluate, Simplify, and Translate Expressions

- **Combine like terms.**
  Step 1. Identify like terms.
  Step 2. Rearrange the expression so like terms are together.
  Step 3. Add the coefficients of the like terms

2.3 Solving Equations Using the Subtraction and Addition Properties of Equality

- **Determine whether a number is a solution to an equation.**
  Step 1. Substitute the number for the variable in the equation.
  Step 2. Simplify the expressions on both sides of the equation.
  Step 3. Determine whether the resulting equation is true. If it is true, the number is a solution.

  If it is not true, the number is not a solution.
- **Subtraction Property of Equality**
  - For any numbers a, b, and c,

if	$a = b$
then	$a - c = b - c$

- **Solve an equation using the Subtraction Property of Equality.**
  Step 1. Use the Subtraction Property of Equality to isolate the variable.
  Step 2. Simplify the expressions on both sides of the equation.
  Step 3. Check the solution.
- **Addition Property of Equality**
  - For any numbers a, b, and c,

if	$a = b$
then	$a + c = b + c$

- **Solve an equation using the Addition Property of Equality.**
  Step 1. Use the Addition Property of Equality to isolate the variable.
  Step 2. Simplify the expressions on both sides of the equation.
  Step 3. Check the solution.

2.4 Find Multiples and Factors

Divisibility Tests	
A number is divisible by	
2	if the last digit is **0, 2, 4, 6,** or **8**
3	if the sum of the digits is divisible by **3**
4	if the last two digits are a number divisible by **4**
5	if the last digit is **5** or **0**
6	if divisible by both **2** and **3**
10	if the last digit is **0**

- **Factors** If $a \cdot b = m$, then a and b are factors of m, and m is the product of a and b.
- **Find all the factors of a counting number.**
  Step 1. Divide the number by each of the counting numbers, in order, until the quotient is smaller than the divisor.
    a. If the quotient is a counting number, the divisor and quotient are a pair of factors.
    b. If the quotient is not a counting number, the divisor is not a factor.
  Step 2. List all the factor pairs.
  Step 3. Write all the factors in order from smallest to largest.
- **Determine if a number is prime.**
  Step 1. Test each of the primes, in order, to see if it is a factor of the number.
  Step 2. Start with 2 and stop when the quotient is smaller than the divisor or when a prime factor is found.
  Step 3. If the number has a prime factor, then it is a composite number. If it has no prime factors, then the number is prime.

2.5 Prime Factorization and the Least Common Multiple

- **Find the prime factorization of a composite number using the tree method.**
  Step 1. Find any factor pair of the given number, and use these numbers to create two branches.
  Step 2. If a factor is prime, that branch is complete. Circle the prime.
  Step 3. If a factor is not prime, write it as the product of a factor pair and continue the process.
  Step 4. Write the composite number as the product of all the circled primes.
- **Find the prime factorization of a composite number using the ladder method.**
  Step 1. Divide the number by the smallest prime.
  Step 2. Continue dividing by that prime until it no longer divides evenly.
  Step 3. Divide by the next prime until it no longer divides evenly.
  Step 4. Continue until the quotient is a prime.
  Step 5. Write the composite number as the product of all the primes on the sides and top of the ladder.
- **Find the LCM by listing multiples.**
  Step 1. List the first several multiples of each number.